UNIVERSITY
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Master's thesis

Topological quantum computing with the Kitaev chain

Solving quantum computing and
providing the world with infinite porridge

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Abstract

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Chapter 2

The Kitaev chain

The Kitaev chain is a toy model of a one dimensional spinless p-wave superconductor, and is an example of material that can realize a topological superconducting phase [1]. It consists of a chain of spinless fermions with nearest-neighbour hopping and superconducting pairing, under the influence of a chemical potential. The Hamiltonian takes the form

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{2.1}$$

where a_j and $a_j^\dagger$ are annihilation and creation operators for the fermionic sites j in the chain. ω is a hopping amplitude, μ is the chemical potential, and $\Delta = |\Delta| e^{i\theta}$ is the induced superconducting gap. The Hamiltonian serves to give rise to Majorana fermion quasiparticles at its boundaries. These Majorana fermions can be used as qubits, and a network of chains can be used for quantum computation.

Majorana fermions are interesting candidates for qubits as their topological nature provides robustness against local perturbations and certain sources of decoherence. Their non-Abelian nature also allows for braiding, where the majorana fermions exchange position and information about the path is stored in the wavefunction.

2.1 Majorana fermions

A Majorana fermion is a particle that is its own antiparticle. The Majorana operators can be defined as

$$c_{2j-1} = a_j + a_j^\dagger, \quad c_{2j} = \frac{a_j - a_j^\dagger}{i}, \quad (j = 1, ..., N) \tag{2.2}$$

which are hermitian, and anticommute, meaning they satisfy the relations

$$c_m^\dagger = c_m, \quad c_l c_m + c_m c_l = 2\delta_{lm}, \quad (l, m = 1, ..., 2N). \tag{2.3}$$

Using the definitions of the majorana operators 2.2 we can find the number operator for a site j

$$a_j^\dagger a_j = \frac{1}{2} \left(1 + i c_{2j-1} c_{2j} \right), \tag{2.4}$$

where the term $ic_{2j-1}c_{2j}$ corresponds to a phase error. If c_{2j-1} and c_{2j} are not on the same site, but separated in space, then $c_{2j-1}c_{2j}$ would represent an interaction

between two separated Majorana sites, requiring the environment to couple the Majorana fermions simultaneously to create the phase error. Physical noise and imperfections are typically local, so placing the Majorana sites far enough away from each other would decrease the risk of a phase error. Further, a single Majorana operator cannot appear by itself in a reasonable Hamiltonian as it would not preserve fermionic parity; flipping the system from an even to odd (or vice versa) number of fermions. As such, the Majorana operators will always come in pairs $c_i c_l$, and thus no parity preserving term can involve just one isolated Majorana. Together this tells us that errors would either have to break parity, or couple both separated Majoranas at once, which is highly unlikely and thus the Majorana fermions are protected from local errors.

2.1.1 Rewriting the Hamiltonian in terms of Majorana operators

We can express the Hamiltonian in terms of Majorana operators. As we will see, this transformation will provide insight about the structure of the phases and the emergence of the edge modes. Before doing so we simplify the calculation by hiding the phase θ in the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}} a_j + e^{-i\frac{\theta}{2}} a_j^\dagger \tag{2.5}$$

$$c_{2j} = -ie^{i\frac{\theta}{2}} a_j + ie^{-i\frac{\theta}{2}} a_j^\dagger. \tag{2.6}$$

We can now perform the transformation of the Hamiltonian by inverting these relations and inserting into the original Hamiltonian 2.1. Together with the hermiticity and anticommutation relations 2.3 we get the Hamiltonian

$$H = \frac{i}{2} \sum_j \left[-\mu c_{2j-1} c_{2j} + (\omega + |\Delta|)\, c_{2j} c_{2j+1} + (-\omega + |\Delta|)\, c_{2j-1} c_{2j+2} \right]. \tag{2.7}$$

This calculation is shown in appendix A.1. Staring at this Hamiltonian, we can first see that it is on the general quadratic Majorana form

$$H = \frac{i}{4} \sum_{l,m} A_{lm} c_l c_m \tag{2.8}$$

with A being a real, antisymmetric matrix. With A being real antisymmetric, it encodes the intrinsic particle-hole symmetry of the superconducting system. And the single particle spectrum is fully determined by the eigenvalues of A, reducing the problem from a many-body operator problem to a single-particle linear algebra problem. This is the basis of BdG, as we will get back to later. Further, we see that all direct mentions of the complex phase θ are gone from H. This reflects that θ can be removed by an appropriate gauge transformation, and is therefore irrelevant for the bulk spectrum and the topological classification of the phases, which we will also touch upon later.

The Majorana representation also gives us a geometric picture of the chain. The Hamiltonian now consists of three terms, providing three different types of couplings. An on-site term $-\mu c_{2j-1} c_{2j}$, and two nearest-neighbour terms $(\omega + |\Delta|) c_{2j} c_{2j+1}$ and $(-\omega + |\Delta|) c_{2j-1} c_{2j+2}$. The on-site term produces two Majorana modes per site j, so for a chain of length L we get $2L$ Majorana operators. We can garner further insight by studying specific cases for the parameters μ, ω and $|\Delta|$ of the Hamiltonian.

Trivial case $\omega = |\Delta| = 0, \mu > 0$

In the case where μ dominates the Hamiltonian reduces to

$$H = -\mu \sum_j \left(a_j^\dagger a_j - \frac{1}{2} \right) = \frac{i}{2} \mu \sum_j c_{2j-1} c_{2j}. \tag{2.9}$$

In this case the system is topologically trivial as the Majoranas on each site are glued together to form a complex fermion. There are no unpaired Majorana fermions at the edges of the chain.

Diagonalization in the $\omega = |\Delta| > 0, \mu = 0$ case

In the case where the hopping and superconducting parameters dominate the Hamiltonian reduces to

$$H = i \sum_j \omega c_{2j} c_{2j+1}, \tag{2.10}$$

where the nearest neighbours between different sites dominate. This is an interesting case as c_2 couples to c_3, c_4 couples to c_5, and so on. But c_1 and $c_2 L$, the Majorana edge states, do not appear in the Hamiltonian. They have however not disappeared physically, but rather exist as zero energy nonlocal fermionic modes at the edges of the wire. This is the most important insight from the transformation of the Hamiltonian as these are the states we were looking to create.

We can now define new annihilation and creation operators for the sites

$$d_j = \frac{1}{2} \left(c_{2j} + i c_{2j+1} \right) \tag{2.11}$$

$$d_j^\dagger = \frac{1}{2} \left(c_{2j} - i c_{2j+1} \right). \tag{2.12}$$

Such that we can rewrite the Hamiltonian

$$H = 2\omega \sum_{j=1}^{N-1} \left(d_j^\dagger d_j - \frac{1}{2} \right) \tag{2.13}$$

Arbitrary ω, μ, Δ

2.2 Topology

2.3 BdG

2.4 Quantum computation & the T-junction

Chapter 5

Spørsmål

Generelle spørsmål: 08.03.26: Bruker jeg bra notasjon eller burde jeg bytte til mer kompakt notasjon (dot for derivert)?

Eivind master:

Kitaev original article https://arxiv.org/pdf/cond-mat/0010440 - Hvordan kommer majorana fermionene til kantene - er det bare et matematisk triks ved å definere hamiltonianen ved riktige operatorer? Ref. figur i Kitaev artikkel

"Introduction to topological superconductivity and Majorana fermions Martin Leijnse and Karsten Flensberg " https://arxiv.org/pdf/1206.1736

Non-Abelian statistics and topological quantum information processing in 1D wire networksfile:///C:/Users/axlkl/Downloads/nphys1915.pdf

From its supplementary: -

Topological Quantum Computation https://arxiv.org/pdf/quant-ph/0101025

5.1 Huskeliste / idemyldring

Recap av Kitaev artikkelen om Majorana fermioner - Ikke bruk Arxiv som kilde - Flesh ut Kitaev kapittel med - Hvorfor operatorene er som de er mtp. hermitisk, antikommuterende - Mer kondenserte språk? - Mer direkte eksempler på hva Majoranaene tåler - Matematiske nøyaktigheter og innsikter

Adiabatisk (betydning)

Kutte kjede - Hvordan kjeden er kuttet; hva som faktisk endres og hva det tilsvarer fysisk

T-junction - Flytte rundt på fermionene. Hvordan de flyttes rundt, helt spesifikt

Sammenligne med Eivind - god kode - Resultater for kutte kjede - Diskusjon mtp. RK4 vs. euler cromer; nøyaktighet, tidssteg, tidsbruk, trunkering osv.

Bibliography

[1] Alexei Kitaev. “Unpaired Majorana Fermions in Quantum Wires”. In: *Physics-Uspekhi* 44.10S (Oct. 2001), pp. 131–136. ISSN: 1468-4780. DOI: 10.1070/1063-7869/44/10S/S29. arXiv: cond-mat/0010440. (Visited on 29/01/2026).

Appendix A

Calculations

A.1 Transforming Kitaev chain Hamiltonian to Majorana operators

This appendix shows how transform the original Kitaev chain Hamiltonian expressed in fermionic operators a_j and $a_j^\dagger$ to being expressed in terms of the majorana operators c_{2j-1} and $c_2 j$.

Starting with the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}} a_j + e^{-i\frac{\theta}{2}} a_j^\dagger \tag{A.1}$$
$$c_{2j} = -ie^{i\frac{\theta}{2}} a_j + ie^{-i\frac{\theta}{2}} a_j^\dagger. \tag{A.2}$$

We can invert them, leading to four expressions

$$a_j = e^{-i\frac{\theta}{2}} c_{2j-1} - e^{-i\theta} a_j^\dagger \tag{A.3}$$
$$= ie^{-i\frac{\theta}{2}} c_{2j} + e^{-i\theta} a_j^\dagger \tag{A.4}$$
$$a_j^\dagger = e^{i\frac{\theta}{2}} c_{2j-1} - e^{i\theta} a_j \tag{A.5}$$
$$= -ie^{i\frac{\theta}{2}} c_{2j} + e^{i\theta} a_j. \tag{A.6}$$

Combining the expressions for a_j, and for $a_j^\dagger$ we find

$$a_j = \frac{1}{2}\left(e^{-i\frac{\theta}{2}} c_{2j-1} - e^{-i\theta} a_j^\dagger + e^{-i\theta} a_j^\dagger + ie^{-i\frac{\theta}{2}} c_2 j\right) \tag{A.7}$$
$$= \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + ic_{2j}\right) \tag{A.8}$$
$$a_j^\dagger = \frac{1}{2}\left(e^{i\frac{\theta}{2}} c_{2j-1} - e^{i\theta} a_j + e^{i\theta} a_j - ie^{i\frac{\theta}{2}} c_{2j}\right) \tag{A.9}$$
$$= \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - ic_{2j}\right). \tag{A.10}$$

We now substitute these expressions into the original Hamiltonian

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{A.11}$$

$$= -\omega \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2(j+1)-1} + i c_{2(j+1)} \right) \tag{A.12}$$

$$- \omega \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2(j+1)-1} - i c_{2(j+1)} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) \tag{A.13}$$

$$- \mu \sum_j \left(\frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) - \frac{1}{2} \right) \tag{A.14}$$

$$+ \Delta \sum_j \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2(j+1)-1} + i c_{2(j+1)} \right) \tag{A.15}$$

$$+ \Delta^* \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2(j+1)-1} - i c_{2(j+1)} \right) \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \tag{A.16}$$

$$= -\frac{\omega}{4} \sum_j \left(c_{2j-1} c_{2j+1} + i c_{2j-1} c_{2j+2} - i c_{2j} c_{2j+1} + c_{2j} c_{2j+2} \right) \tag{A.17}$$

$$- \frac{\omega}{4} \sum_j \left(c_{2j+1} c_{2j-1} + i c_{2j+1} c_{2j} - i c_{2j+2} c_{2j-1} + c_{2j+2} c_{2j} \right) \tag{A.18}$$

$$- \frac{\mu}{4} \sum_j \left(c_{2j-1}^2 + i c_{2j-1} c_{2j} - i c_{2j} c_{2j-1} + c_{2j}^2 - 2 \right) \tag{A.19}$$

$$+ \frac{|\Delta|}{4} e^{i\theta} \sum_j e^{-i\theta} \left(c_{2j-1} c_{2j+1} + i c_{2j-1} c_{2j+2} + i c_{2j} c_{2j+1} - c_{2j} c_{2j+2} \right) \tag{A.20}$$

$$+ \frac{|\Delta|}{4} e^{-i\theta} \sum_j e^{i\theta} \left(c_{2j+1} c_{2j-1} - i c_{2j+1} c_{2j} - i c_{2j+2} c_{2j-1} - c_{2j+2} c_{2j} \right) \tag{A.21}$$

$$= -\frac{\omega}{4} \sum_j \left(0 + 2 i c_{2j-1} c_{2j+2} - 2 i c_{2j} c_{2j+1} + 0 \right) \tag{A.22}$$

$$- \frac{\mu}{4} \sum_j \left(2 i c_{2j-1} c_{2j} + 2 - 2 \right) \tag{A.23}$$

$$+ \frac{|\Delta|}{4} \sum_j \left(0 + 2 i c_{2j-1} c_{2j+2} + 2 i c_{2j} c_{2j+1} + 0 \right) \tag{A.24}$$

$$= \frac{i}{2} \sum_j \left[-\mu c_{2j-1} c_{2j} + \left(\omega + |\Delta| \right) c_{2j} c_{2j+1} + \left(-\omega + |\Delta| \right) c_{2j-1} c_{2j+2} \right] \tag{A.25}$$

Where we used $\Delta = |\Delta| e^{i\theta}$ and the anticommutation relations

$$\{ c_l, c_m \} = c_l c_m + c_m c_l = 2 \delta_{lm} \tag{A.26}$$

$$c_l c_m = -c_m c_l (m \neq l) \tag{A.27}$$

$$c_l^2 = 1. \tag{A.28}$$